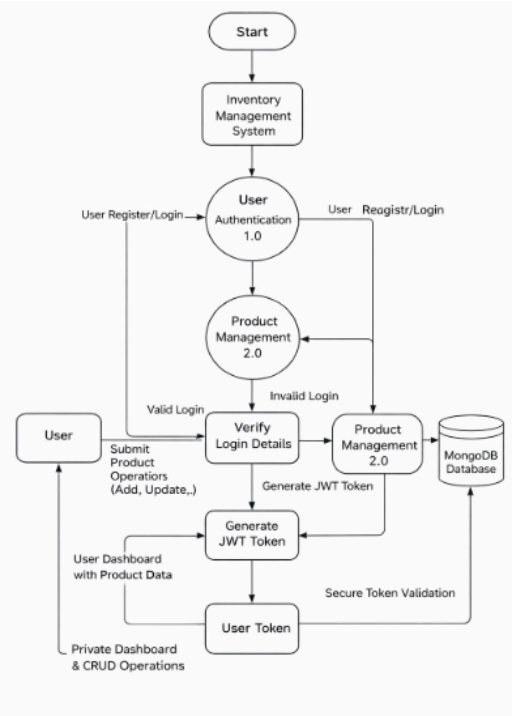
Project Design Phase-II
Data Flow Diagram & User Stories

Date	07/03/2026
Team ID	SWTID-2026-6411
Project Name	product inventory management
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.



User Stories

Use the below template to list all the user stories for the product.

User Type	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web User)	USN-1	As a user, I can register and log in securely using JWT authentication	User can create account and receive token after login	High	Sprint-1
Customer (Web User)	USN-2	As a user, I can add new products to my inventory	Product is saved and visible only to logged-in user	High	Sprint-1
Customer (Web User)	USN-3	As a user, I can view all my products in a dashboard	Products load successfully with valid token	High	Sprint-1
Customer (Web User)	USN-4	As a user, I can update product details like price and quantity	Changes are saved and reflected instantly	Medium	Sprint-2
Customer (Web User)	USN-5	As a user, I can delete products from my inventory	Product disappears from dashboard after deletion	Medium	Sprint-2
Customer (Web User)	USN-6	As a user, I can search products by name	Filtered results appear based on keyword	Low	Sprint-2

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

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Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Authentication & Authorization**	User can register with email and password
		User can log in securely using JWT
		Only authenticated users can access dashboard
FR-2	Product Management (CRUD)	Add new product with name, price, quantity and description
		View all products in dashboard

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should have a simple and intuitive user interface so users can manage products easily.
NFR-2	Security	JWT authentication must be used. Passwords should be hashed and unauthorized access must be prevented.

NFR-3	Reliability	The system should provide consistent performance with minimal server errors or downtime.
NFR-4	Performance	Product data should load quickly and API responses should be returned within 2 seconds.
NFR-5	Availability	The system should remain accessible to users with minimal downtime during operation.
NFR-6	Scalability	The architecture should support increasing numbers of users and products without performance issues. Future features should be easy to integrate.

Project Design Phase-II
Technology Stack (Architecture & Stack)

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Technical Architecture:

The system follows a MERN Stack Architecture where the React frontend communicates with a Node.js + Express backend through REST APIs. Authentication is handled using JWT tokens, and all user and product data is stored in MongoDB.

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	Application Logic-1	User authentication, JWT token generation and validation	Node.js, Express.js, bcryptjs, jsonwebtoken
2	Application Logic-2	Product CRUD operations and search functionality	Node.js, Express.js REST API
3	Database	Store user accounts and product inventory data securely	MongoDB Atlas, Mongoose
4	API Communication	Handles secure data transfer between frontend and backend	REST API, Axios, HTTPS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Backend frameworks used in the system	Node.js, Express.js, Mongoose
2	Scalable Architecture	Modular MVC architecture supporting future features	REST API Architecture, JWT Middleware
3	Security	Secure authentication and protected routes	JWT Authentication, bcrypt Password Hashing
4	Performance Optimization	Fast loading UI and efficient database queries	Vite, MongoDB Indexing
5	Responsive Design	Supports multiple devices and screen sizes	Tailwind CSS, Responsive Layout